

Exercises for Chapter 1

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June 16, 2025

Unless the contrary is explicitly stated, all numbers that are mentioned in these exercises are understood to be real.

1. If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx are irrational.

Proof. If $r + x$ is rational, then $r + x = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$. Then

$$\begin{aligned}(r + x) * b &= a \\ \implies rb + xb &= a \\ \implies x &= \frac{a - rb}{b},\end{aligned}$$

so that x is rational, contradicting the hypothesis.

Similarly, if rx is rational, then again $rx = \frac{a}{b}$ for $a, b \in \mathbb{Z}$, and

$$x = \frac{a}{br} \in \mathbb{Q}.$$

□

2. Prove that there is no rational number whose square is 12.

Proof. Assume that there exists $a, b \in \mathbb{Z}$ coprime such that $\left(\frac{a}{b}\right)^2 = 12$. Then

$$\begin{aligned}\frac{a^2}{b^2} &= 12 \\ \implies a^2 &= 12b^2.\end{aligned}$$

Thus 3 divides a^2 , so 3 divides a and in fact 9 divides a^2 . Since 3 only divides 12 once, it must then divide b^2 and thus b , contradicting our assumption of a and b being coprime. □

3. The axioms for multiplication imply the following statements.

- (a) If $x \neq 0$ and $xy = xz$ then $y = z$.
- (b) If $x \neq 0$ and $xy = x$ then $y = 1$.
- (c) If $x \neq 0$ and $xy = 1$ then $y = 1/x$.

(d) If $x \neq 0$ then $1/(1/x) = x$.

Proof.

(a) Since $x \neq 0$, x has an inverse $1/x$, and

$$\begin{aligned} xy = xz &\implies 1/x(xy) = 1/x(xz) \\ &\implies ((1/x)x)y = ((1/x)x)z \\ &\implies 1y = 1z \\ &\implies y = z. \end{aligned}$$

(b) Take (a) with $z = 1$.

(c) Since $x \neq 0$, x has an inverse $1/x$, and

$$\begin{aligned} 1/x(xy) &= (1/x) * 1 \\ \implies ((1/x)x)y &= 1/x \\ \implies 1 * y &= 1/x \\ \implies y &= 1/x. \end{aligned}$$

(d) Indeed, $(1/x) * x = 1$ by definition.

□

4. Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E . Prove that $\alpha \leq \beta$.

Proof. Given $x \in E$, by definition $\alpha \leq x$ and $x \leq \beta$. By transitivity, $\alpha \leq \beta$.

□

5. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Let $\alpha = -\sup(-A)$. Then

$$\begin{aligned} -\alpha &\geq a \quad \forall a \in -A \\ \implies \alpha &\leq -a, \end{aligned}$$

so α is a lower bound of $-A$. Given $\varepsilon \geq 0$, there exists $b \in A$ such that $-\alpha - \varepsilon < b$, so $\alpha + \varepsilon > -b$. This implies that α is in fact the greatest lower bound of A , since if β were a greater lower bound, we could take $\varepsilon = \beta - \alpha > 0$ and have $\alpha + \varepsilon = \beta \leq a$ for all $a \in A$, contradicting the above.

□

6. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t , where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x .

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y .

Proof.

(a) Note that $m = rn$ and $p = rq$. So

$$\begin{aligned} (b^m)^{1/n} &= (b^{rn})^{1/n} \\ &= ((b^r)^n)^{1/n} \\ &= b^r \\ &= ((b^r)^q)^{1/q} \\ &= (b^{rq})^{1/q} \\ &= (b^p)^{1/q}. \end{aligned}$$

(b) Let $r = a/b$ and $s = c/d$ for $a, b, c, d \in \mathbb{Z}$. Then

$$\begin{aligned} b^{r+s} &= b^{a/b+c/d} \\ &= b^{(ad+bc)/(bd)} \\ &= (b^{ad+bc})^{1/(bd)} \\ &= (b^{ad} b^{bc})^{1/(bd)} \\ &= (b^{ad})^{1/(bd)} (b^{bc})^{1/(bd)} \\ &= b^{(ad)/(bd)} b^{(bc)/(bd)} \\ &= b^{a/b} b^{c/d} \\ &= b^r b^s. \end{aligned}$$

(c) Let $t \leq r$ and $t \in \mathbb{Q}$. Since $b > 1$ and $t - r \leq 0$

$$\begin{aligned} b^{t-r} &\leq 1 \\ \implies b^r b^{t-r} &\leq b^r \\ \implies b^{r+t-r} &\leq b^r \\ \implies b^t &\leq b^r. \end{aligned}$$

Thus b^r is an upper bound of $B(r)$, and since $b^r \in B(r)$, $b^r = \sup B(r)$.

(d) Given $x, y \in \mathbb{R}$, $b^{x+y} = \sup B(x+y)$.

$$\begin{aligned} &= (\sup B(x)) (\sup B(y)) \\ &= b^x b^y \end{aligned}$$

INCOMPLETE

□

7. Fix $b > 1$, $y > 0$, and prove that there is a unique real x such that $b^x = y$, by completing the following outline. (This x is called the *logarithm of y to the base b* .)

- (a) For any positive integer n , $b^n - 1 \geq n(b - 1)$.
- (b) Hence $b - 1 \geq n(b^{1/n} - 1)$.
- (c) If $t > 1$ and $n > (b - 1)/(t - 1)$, then $b^{1/n} < t$.
- (d) If w is such that $b^w < y$, then $b^{w+(1/n)} < y$ for sufficiently large n ; to see this, apply part (c) with $t = y \cdot b^{-w}$.
- (e) If $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n .
- (f) Let A be the set of all w such that $b^w < y$, and show that $x = \sup A$ satisfies $b^x = y$.
- (g) Prove that this x is unique.

Proof. (a) For $n = 1$, $b^1 - 1 = 1(b - 1)$, so the inequality holds. We proceed by induction, assuming the inequality holds for n . Then

$$\begin{aligned} b^n - 1 &\geq n(b - 1) \\ \implies b(b^n - 1) &\geq n(b - 1) \\ \implies b^{n+1} - b &\geq n(b - 1) \\ \implies b^{n+1} &\geq n(b - 1) + b \\ \implies b^{n+1} - 1 &\geq n(b - 1) + b - 1 \\ \implies b^{n+1} - 1 &\geq (n + 1)(b - 1) \end{aligned}$$

as desired.

(b) WTF Walter

(c) Indeed,

$$\begin{aligned} n(b^{1/n} - 1) &\leq b - 1 < n(t - 1) \\ \implies b^{1/n} - 1 &< t - 1 \\ \implies b^{1/n} &< t. \end{aligned}$$

(d) As suggested, take $t = y \cdot b^{-w}$ since $b^w < y \implies 1 < yb^{-w}$. Then

$$\begin{aligned} b^{1/n} &< y \cdot b^{-w} \\ \implies b^{w+1/n} &< \end{aligned}$$

□

8. Prove that no order can be defined in the complex field that turns it into an ordered field. Hint: -1 is a square.

Proof. We will assume Proposition 1.18 throughout this proof. Since $i * i = -1 < 0$, i must be negative if $\mathbb{C}$ is to have any chance of being an ordered field. Thus $-i > 0$. However, $(-i) * (-i) = -1 < 0$, which contradicts the requirement that positive elements have a positive product. □

9. Suppose $z = a + bi$, $w = c + di$. Define $z < w$ if $a < c$, and also if $a = c$ but $b < d$. Prove that this turns the set of all complex numbers into an ordered set. (This type of order relation is called a *dictionary order*, or *lexicographic order*, for obvious reasons.) Does this ordered set have the least-upper-bound property?

Proof. Given z and w as defined in the statement of the exercise, if $z \neq w$, we need to show that either $z < w$ or $w < z$. Indeed, if $a < c$ or $c < a$, then $z < w$ or $w < z$ respectively. If $a = c$, then either $b < d$ or $d < b$, since if $b = d$, $z = w$. Then we have once again that $z < w$ or $w < z$, respectively. Note that the above relies on the total ordering of the real numbers to be able to compare a with c and b with d .

Consider the set $A = \{1 + bi : b \in \mathbb{R}\}$. Any complex number $a + bi$ with $a > 1$ is an upper bound of A , but

$$1 + bi < a + (b - 1)i < a + bi.$$

proving that there is no least upper bound of A . □

10. Suppose $z = a + bi$, $w = u + iv$, and

$$a = \left(\frac{|w|+u}{2}\right)^{1/2}, \quad b = \left(\frac{|w|-u}{2}\right)^{1/2}.$$

Prove that $z^2 = w$ if $v \geq 0$ and that $(\bar{z})^2 = w$ if $v \leq 0$. Conclude that every complex number (with one exception!) has two complex square roots.

Proof. We verify:

$$\begin{aligned}
z^2 &= \left(\left(\frac{|w|+u}{2} \right)^{1/2} + \left(\frac{|w|-u}{2} \right)^{1/2} i \right)^2 \\
&= \frac{|w|+u}{2} - \frac{|w|-u}{2} + 2 \left(\frac{|w|+u}{2} \right)^{1/2} \left(\frac{|w|-u}{2} \right)^{1/2} i \\
&= u + 2 \left(\frac{|w|^2 - u^2}{4} \right)^{1/2} i \\
&= u + 2 \frac{(u^2 + v^2 - u^2)^{1/2}}{2} i \\
&= u + 2 \frac{v}{2} i \\
&= u + vi \\
&= w.
\end{aligned}$$

Since $(-x)^2 = (-1)^2 x^2 = x^2$ for all elements x of a field, we conclude that each complex number w has two complex square roots z and $-z$ (using the notation from this exercise), except when $z = -z$, in which case $z = w = 0$. $\square$

11. If z is a complex number, prove that there exists an $r \geq 0$ and a complex number w with $|w| = 1$ such that $z = rw$. Are w and r always uniquely determined by z ?

Proof. Let $z = a + bi$. Then setting $w = z/|z| = a/|z| + (b/|z|)i$, clearly $|w| = 1$, $r = |z| \geq 0$, and $|z|w = z$. If $z = 0$, then any w with modulus 1 will work with $r = 0$. $\square$

12. If $z_1, \dots, z_n$ are complex, prove that

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|.$$

Proof. For each k , write $z_k = a_k + b_k i$. Then

$$\begin{aligned}
|z_1 + z_2 + \dots + z_n| &= |(a_1 + a_2 + \dots + a_n) + (b_1 + b_2 + \dots + b_n)i| \\
&= (a_1 + a_2 + \dots + a_n)^2 + (b_1 + b_2 + \dots + b_n)^2 \\
&\leq a_1^2 + a_2^2 + \dots + a_n^2 + b_1^2 + b_2^2 + \dots + b_n^2 \\
&= (a_1^2 + b_1^2) + (a_2^2 + b_2^2) + \dots + (a_n^2 + b_n^2) \\
&= |z_1|^2 + |z_2|^2 + \dots + |z_n|^2
\end{aligned}$$

where the inequality comes directly from the triangle inequality. $\square$

13. If x, y are complex, prove that

$$||x| - |y|| \leq |x - y|.$$

Proof. Let $x = a + bi$ and $y = c + di$. Then

$$\begin{aligned}
||x| - |y|| &= \left| \sqrt{a^2 + b^2} - \sqrt{c^2 + d^2} \right| \\
&= \sqrt{\left(\sqrt{a^2 + b^2} - \sqrt{c^2 + d^2} \right)^2} \\
&= \sqrt{a^2 + b^2 + c^2 + d^2 - 2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2}} \\
&= \sqrt{a^2 + b^2 + c^2 + d^2 - 2\sqrt{(a^2 + b^2)(c^2 + d^2)}} \\
&= \sqrt{a^2 + b^2 + c^2 + d^2 - 2\sqrt{a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2}} \\
&\leq \dots \textbf{UNFINISHED} \\
&= \sqrt{a^2 + b^2 + c^2 + d^2 - 2(ac + bd)} \\
&= \sqrt{a^2 - 2ac + c^2 + b^2 - 2bd + d^2} \\
&= \sqrt{(a - c)^2 + (b - d)^2} \\
&= |(a - c) + (b - d)i| \\
&= |x - y|
\end{aligned}$$

□

14. If z is a complex number such that $|z| = 1$, that is, such that $z\bar{z} = 1$, compute

$$|1 + z|^2 + |1 - z|^2.$$

Proof. Let $z = a + bi$.

$$\begin{aligned}
|1 + z|^2 + |1 - z|^2 &= |(1 + a) + bi|^2 + |(1 - a) - bi|^2 \\
&= (1 + a)^2 + b^2 + (1 - a)^2 + b^2 \\
&= 1 + 2a + a^2 + b^2 + 1 - 2a + a^2 + b^2 \\
&= 2 + 2a^2 + 2b^2 \\
&= 2(1 + a^2 + b^2) \\
&= 2(1 + |a + bi|^2) \\
&= 2 + 2|z|^2 \\
&= 4
\end{aligned}$$

□

15. Under what condition does equality hold in the Schwarz inequality? Schwarz inequality: If $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are complex numbers, then

$$\left| \sum_{j=1}^n a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^n |a_j|^2 \sum_{j=1}^n |b_j|^2.$$

16. Suppose $k \geq 3$, $x, y \in \mathbb{R}^k$, $|x - y| = d > 0$, and $r > 0$. Prove:

(a) If $2r > d$, there are infinitely many $z \in \mathbb{R}^k$ such that

$$|z - x| = |z - y| = r.$$

(b) If $2r = d$, there is exactly one such z .

(c) If $2r < d$, there is no such z .

How must these statements be modified if k is 2 or 1?

Proof.

□

17. Prove that

$$|x + y|^2 + |x - y|^2 = 2|x|^2 + 2|y|^2$$

if $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^k$. Interpret this geometrically, as a statement about parallelograms.

18. If $k \geq 2$ and $x \in \mathbb{R}^k$, prove that there exists $y \in \mathbb{R}^k$ such that $y \neq 0$ but $x \cdot y = 0$. Is this also true if $k = 1$?

19. Suppose $a \in \mathbb{R}^k$, $b \in \mathbb{R}^k$. Find $c \in \mathbb{R}^k$ and $r > 0$ such that

$$|x - a| = 2|x - b|$$

if and only if $|x - c| = r$. (Solution provided)

20. With reference to the Appendix, suppose that property (III) were omitted from the definition of a cut. Keep the same definitions of order and addition. Show that the resulting ordered set has the least-upper-bound property, that addition satisfies axioms (A1) to (A4) (with a slightly different zero-element!) but that (A5) fails.